

Designing a Secure Network Topology (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 03

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Time on Task:

8 hours, 30 minutes

Progress:

100%

Report Generated: Thursday, June 16, 2022 at 8:03 PM

Section 1: Hands-On Demonstration

Part 1: Design a Simple Network Topology

34. Make a screen capture showing the results of the ping attempt from PC4 to PC3.

The screenshot shows a SolarWinds vWorkstation terminal window with the following content:

```
Designing a Secure Network Topology (3e)
Welcome to Virtual PC Simulator, version 0.8.1
Dedicated to Daling.
Build time: Apr 10 2019 16:35:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.
Executing the startup file

PC4> ip 172.30.1.4/24 172.30.1.254
Checking for duplicate address...
PC4 : 172.30.1.4 255.255.255.0 gateway 172.30.1.254

PC4> save
Saving startup configuration to startup.vpc
. done

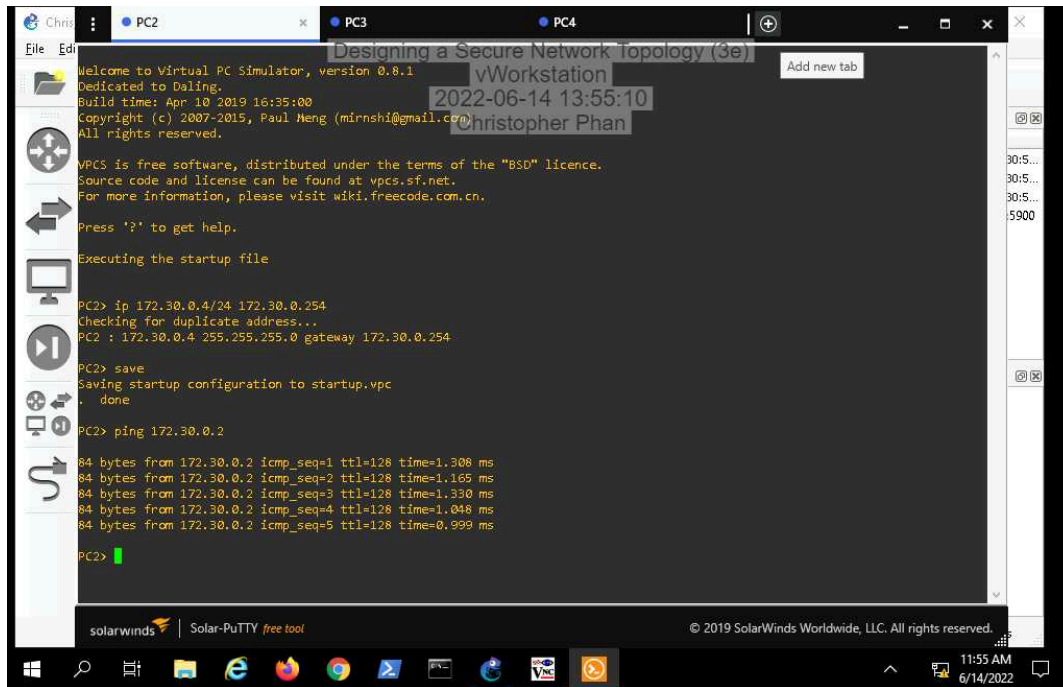
PC4> ping 172.30.1.2

84 bytes from 172.30.1.2 icmp_seq=1 ttl=64 time=0.770 ms
84 bytes from 172.30.1.2 icmp_seq=2 ttl=64 time=1.106 ms
84 bytes from 172.30.1.2 icmp_seq=3 ttl=64 time=0.732 ms
84 bytes from 172.30.1.2 icmp_seq=4 ttl=64 time=0.768 ms
84 bytes from 172.30.1.2 icmp_seq=5 ttl=64 time=1.297 ms

PC4>
```

The terminal window is titled "vWorkstation" and shows the IP configuration and successful ping results for PC4 to PC3 (172.30.1.2).

37. Make a screen capture showing the results of the ping attempt from PC2 to the vWorkstation (172.30.0.2).



49. Make a screen capture showing the IP address assignments in the pfSense console.



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72. Make a screen capture showing the successful ping from PC4 to the vWorkstation.

```
PC4> ping 172.30.0.2
host (172.30.0.2) not reachable

PC4> ip 172.30.1.4/24 172.30.1.254
Checking for duplicate address...
PC4 : 172.30.1.4 255.255.255.0 gateway 172.30.1.254

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> ping 172.30.1.2
84 bytes from 172.30.1.2 icmp_seq=1 ttl=64 time=0.946 ms
84 bytes from 172.30.1.2 icmp_seq=2 ttl=64 time=0.827 ms
84 bytes from 172.30.1.2 icmp_seq=3 ttl=64 time=0.898 ms
84 bytes from 172.30.1.2 icmp_seq=4 ttl=64 time=0.867 ms
84 bytes from 172.30.1.2 icmp_seq=5 ttl=64 time=0.911 ms

PC4> ping 172.30.0.2
84 bytes from 172.30.0.2 icmp_seq=1 ttl=127 time=3.767 ms
84 bytes from 172.30.0.2 icmp_seq=2 ttl=127 time=3.732 ms
84 bytes from 172.30.0.2 icmp_seq=3 ttl=127 time=1.560 ms
84 bytes from 172.30.0.2 icmp_seq=4 ttl=127 time=3.468 ms
84 bytes from 172.30.0.2 icmp_seq=5 ttl=127 time=4.312 ms

PC4>
```

Part 2: Capture Network Traffic to Validate Connectivity

7. Make a screen capture showing the Wireshark capture for the PC2 to PC3 ping request.

Capturing from - [pfSense-Router] em1 to Switch1 Ethernet0

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vWorkstation
2022-06-15 11:07:40
Christopher Phan

No.	Time	Source	Destination	Protocol	Length	Info
12	21.352383	172.30.1.2	172.30.0.4	ICMP	98	Echo (ping) reply id...
13	22.353543	172.30.0.4	172.30.1.2	ICMP	98	Echo (ping) request id...
14	22.356619	172.30.1.2	172.30.0.4	ICMP	98	Echo (ping) reply id...
15	49.654572	172.30.0.2	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
16	50.656805	172.30.0.2	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
17	51.656843	172.30.0.2	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
18	52.657894	172.30.0.2	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1

Protocol type: IPv4 (0x0600)
Hardware size: 6
Protocol size: 4
Opcode: request (1)
Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
Sender IP address: 172.30.0.4
Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
Target IP address: 172.30.0.2

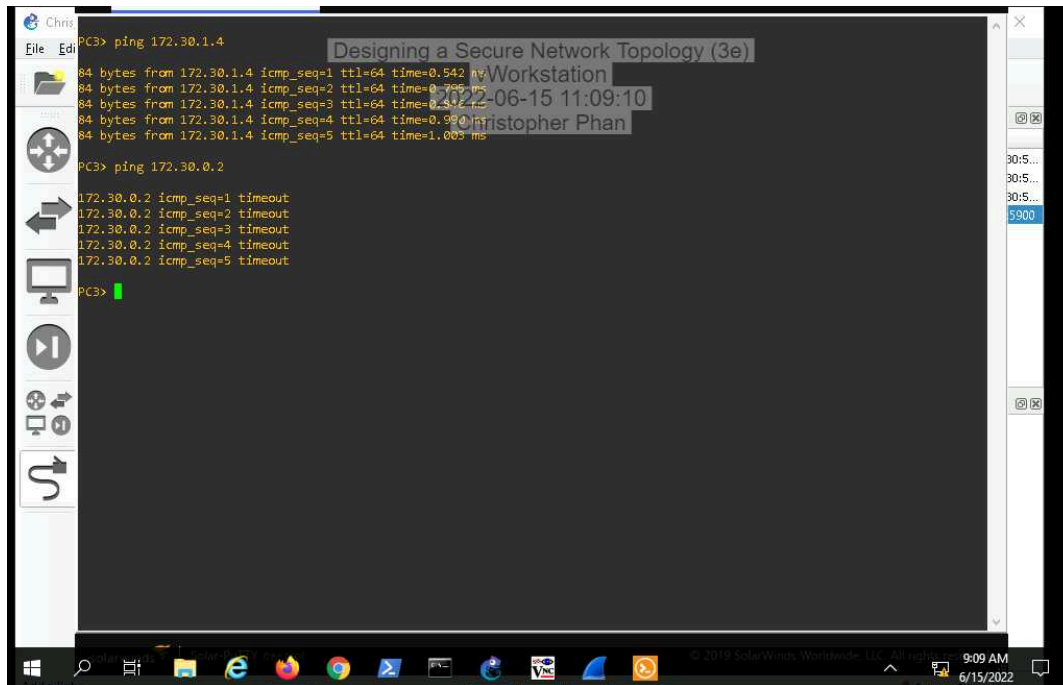
0000 ff ff ff ff ff ff 00 50 79 66 68 01 08 06 00 01P yfh....
0010 08 00 06 04 00 01 00 50 79 66 68 01 ac 1e 00 04P yfh....
0020 ff ff ff ff ff ff ac 1e 00 02 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Target MAC address (arp.dst.hw_mac), 6 bytes | Packets: 18 · Displayed: 18 (100.0%) | Profile: Default

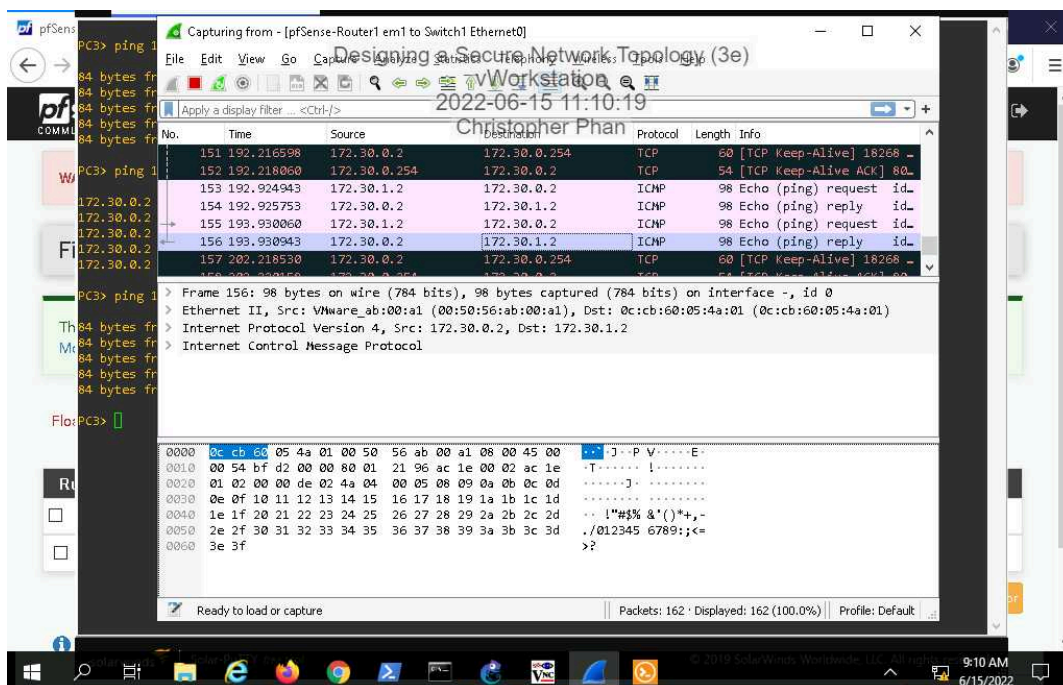
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17. Make a screen capture showing the timed out ping request.



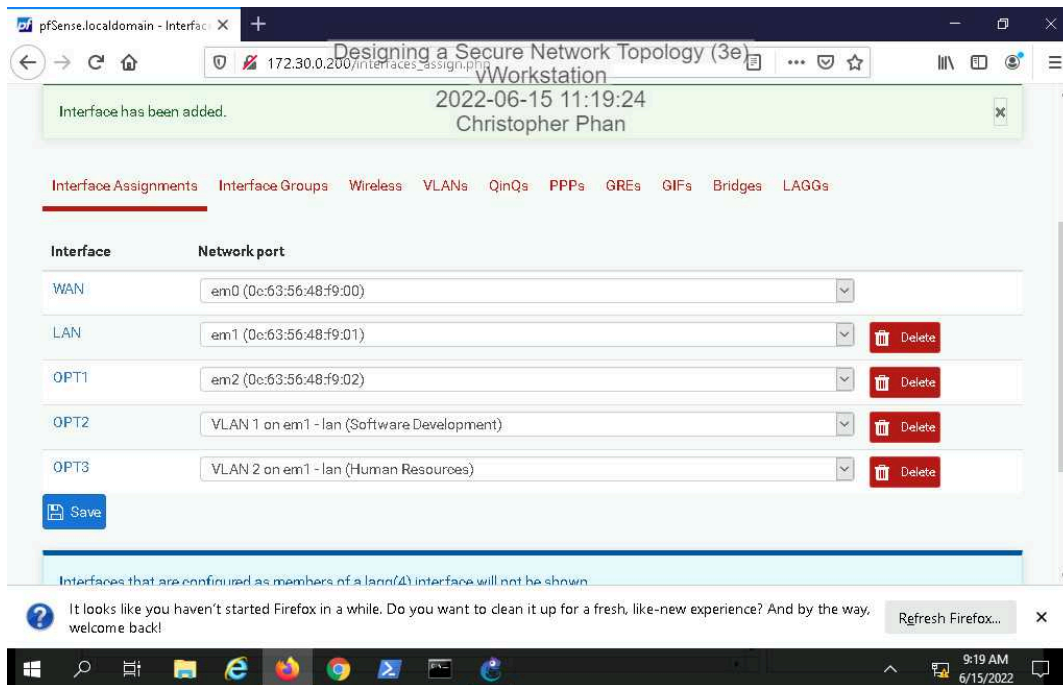
20. Make a screen capture showing the Wireshark capture for the PC3 to vWorkstation ping request.



Section 2: Applied Learning

Part 1: Design a More Complex Network Topology

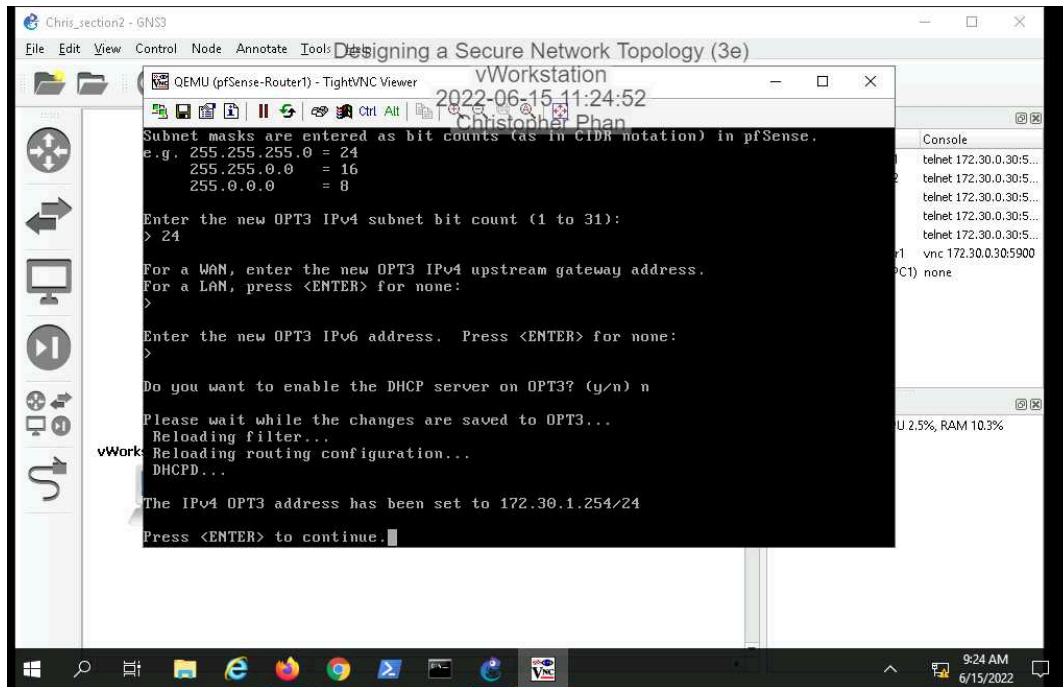
14. Make a screen capture showing the **Interface Assignments** page.



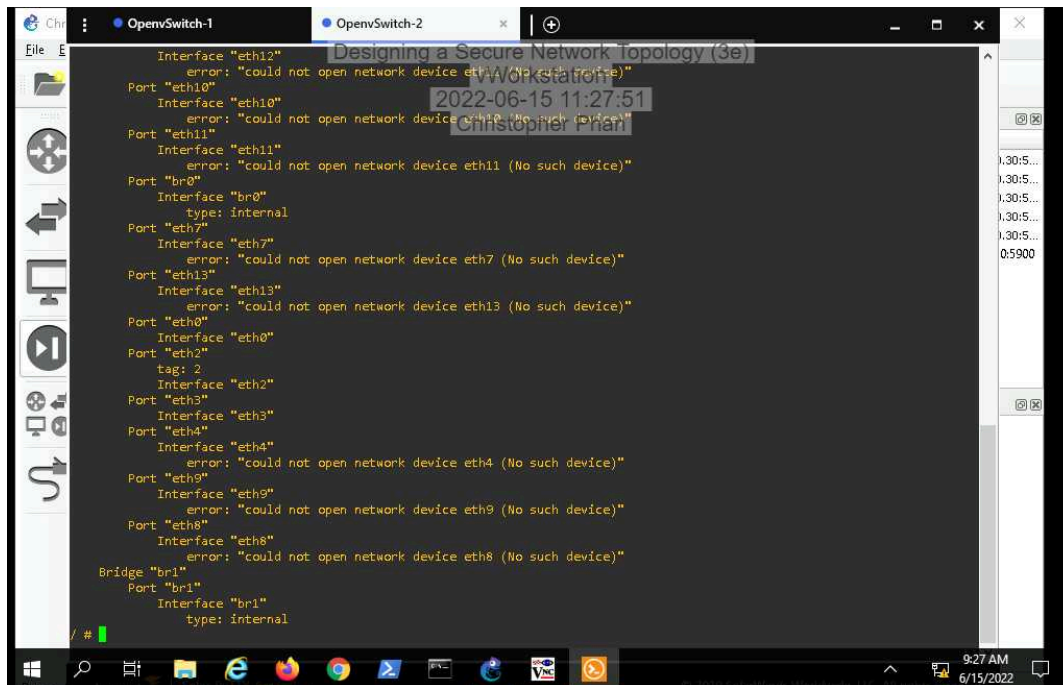
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28. Make a screen capture showing the **updated pfSense router settings with the VLAN IP address assignments.**



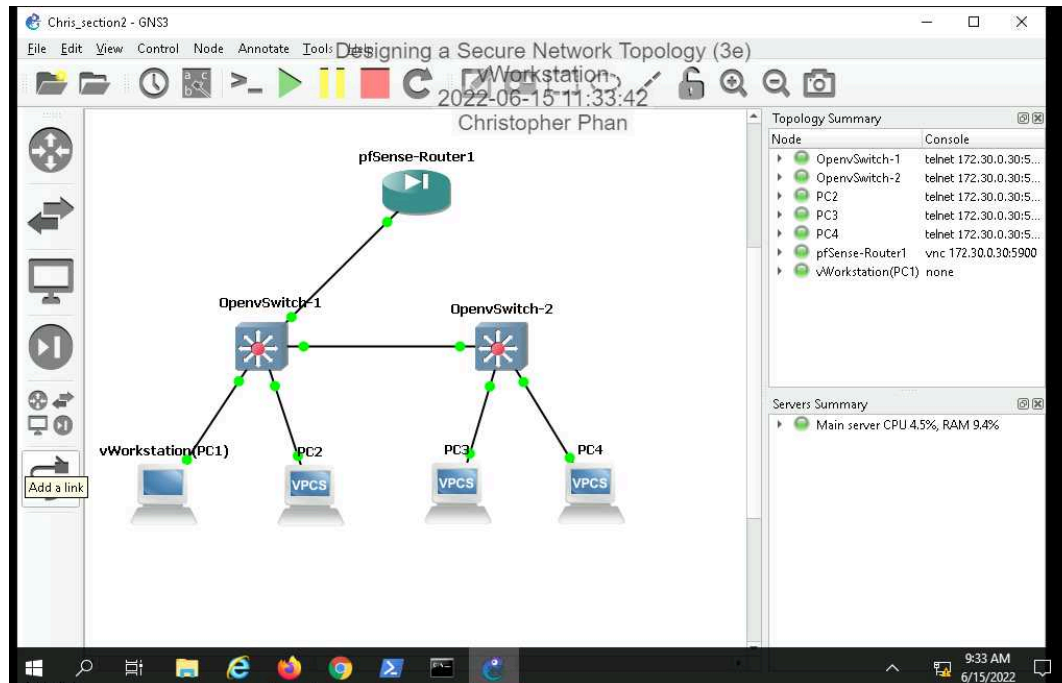
36. Make a screen capture showing the **eth1 and eth2 interfaces with their assigned VLAN tags on Switch2.**



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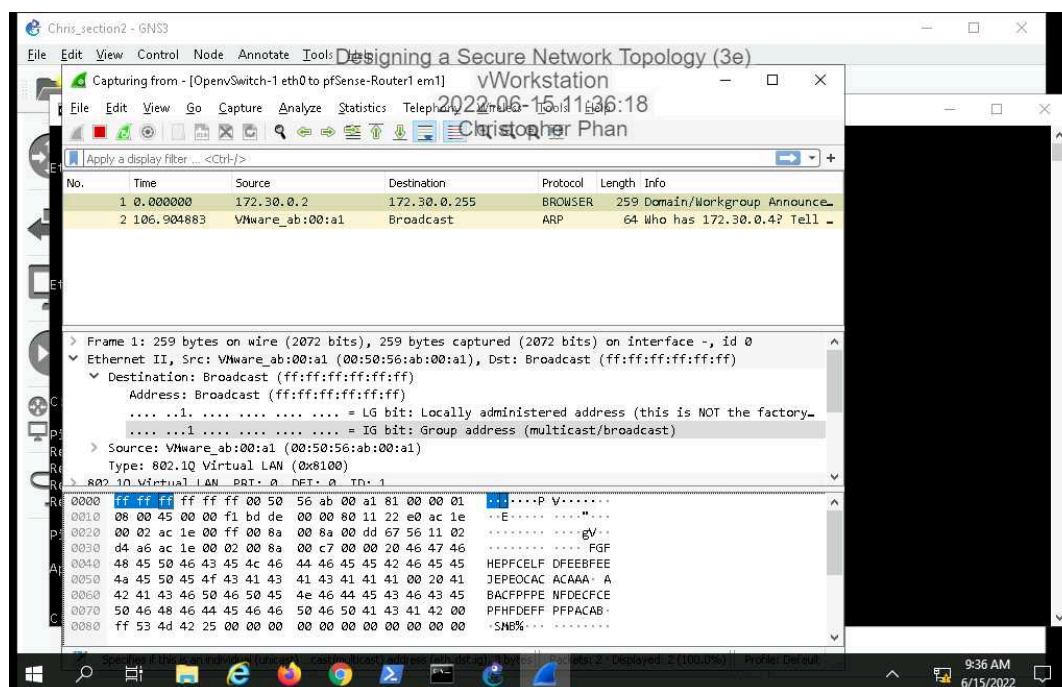
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39. **Make a screen capture** showing the **completed topology**.



Part 2: Capture Network Traffic to Validate Connectivity

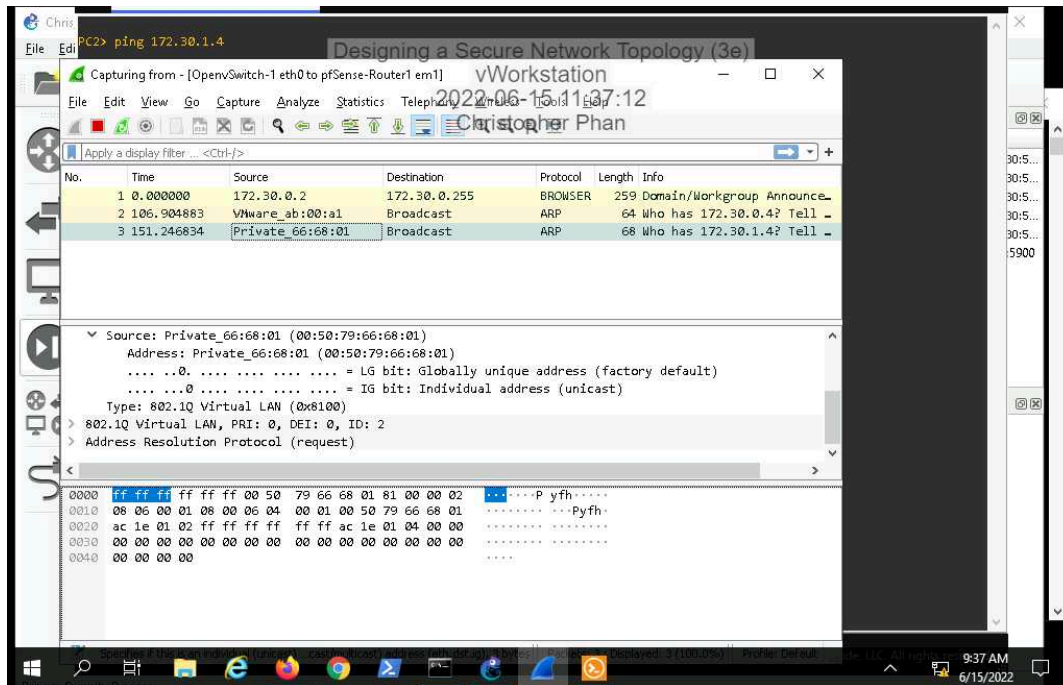
3. **Make a screen capture** showing the **Wireshark capture** for the **vWorkstation to PC3 ping request**.



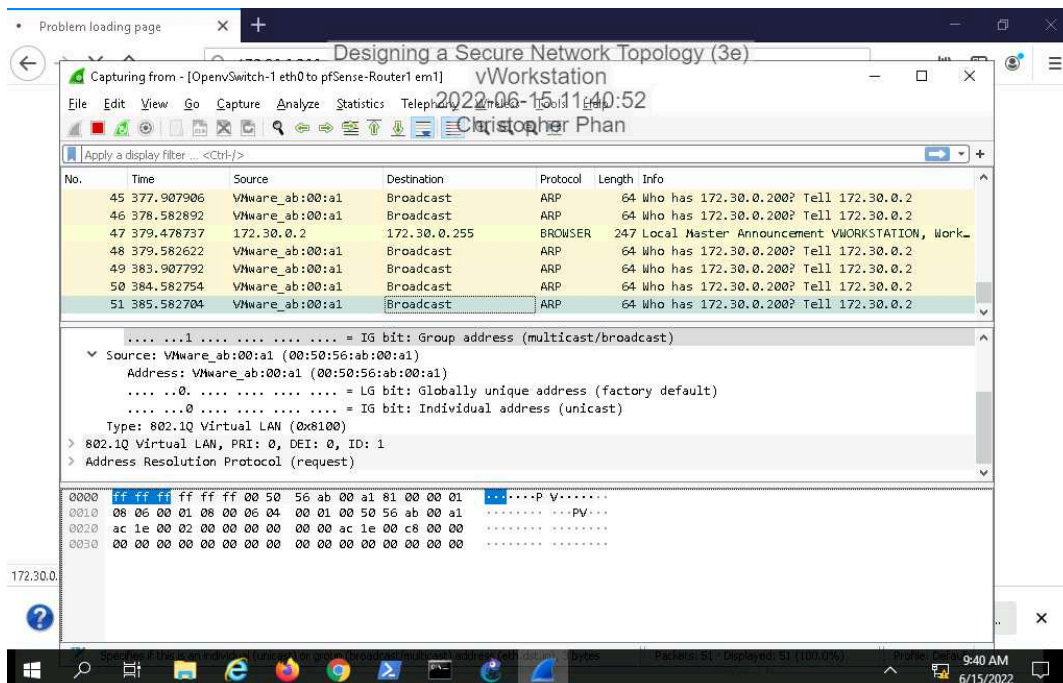
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5. Make a screen capture showing the Wireshark capture for the PC2 to PC4 ping request.



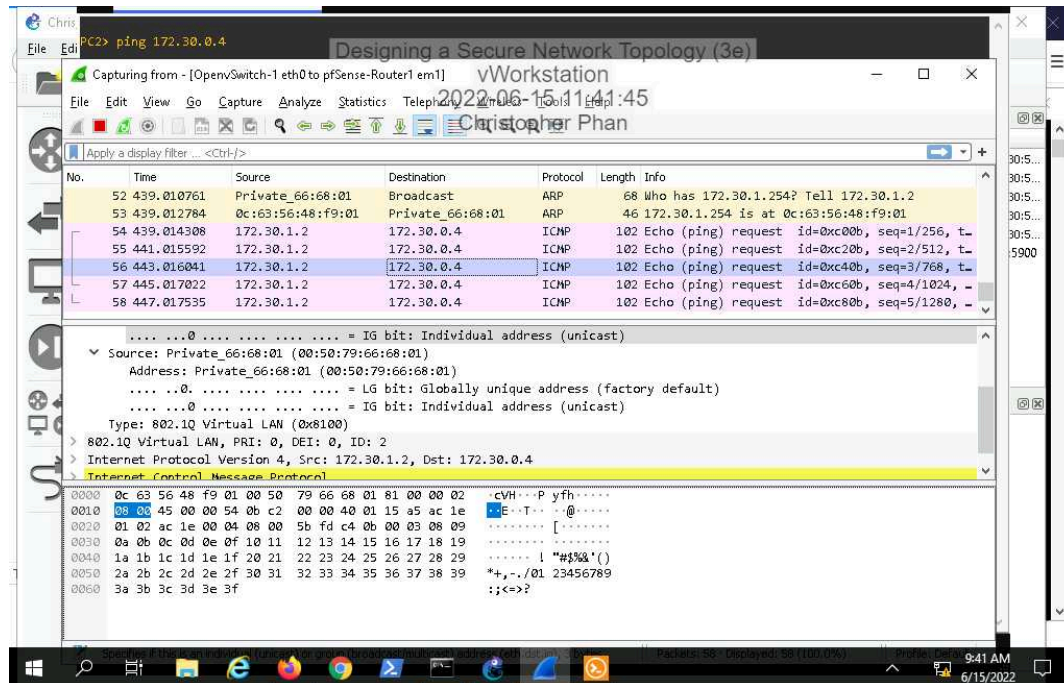
7. Make a screen capture showing the Wireshark capture for the vWorkstation to PC2 ping request.



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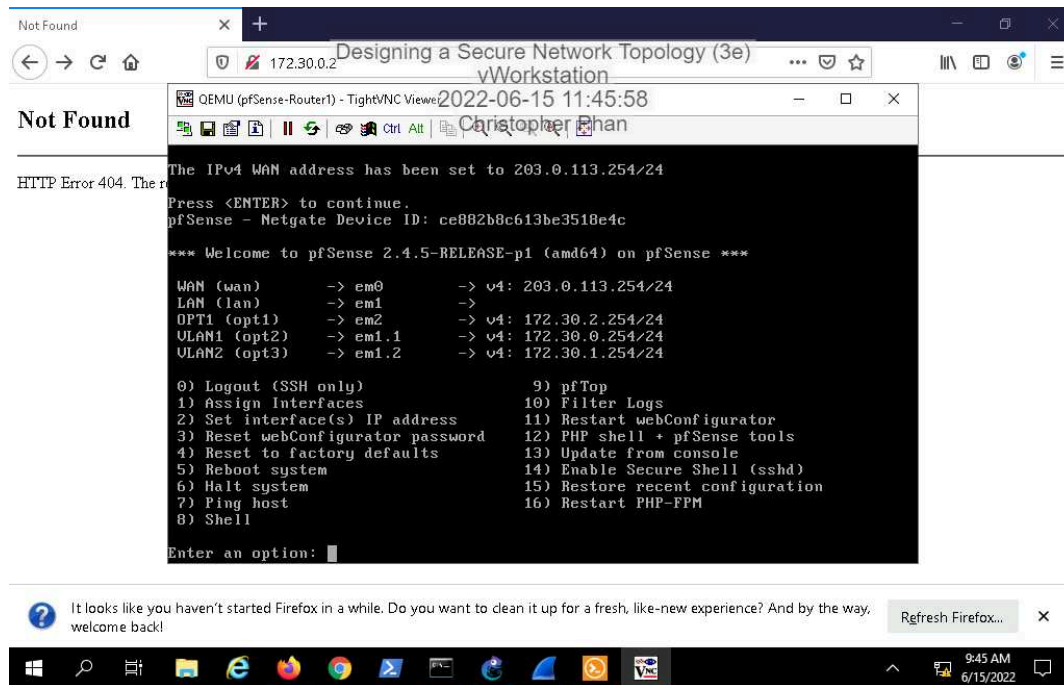
9. Make a screen capture showing the Wireshark capture for the PC2 to PC3 ping request.



Section 3: Challenge and Analysis

Part 1: Enhance the Network Topology with a DMZ

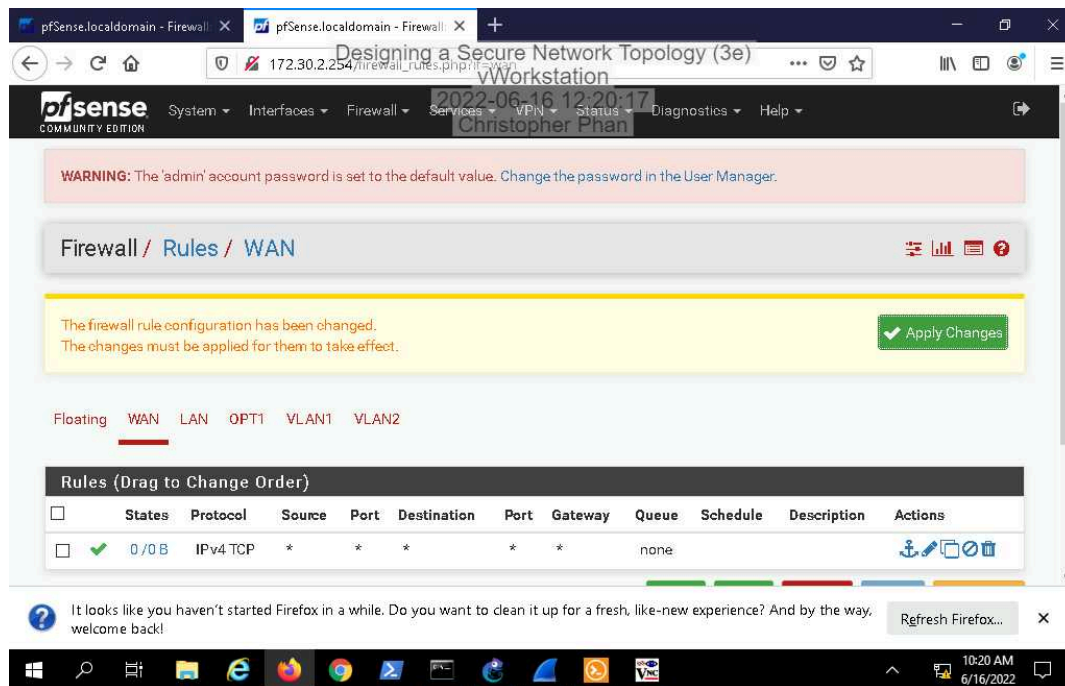
Make a screen capture showing the interface configurations in the pfSense console.



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Make a screen capture showing the firewall rule on the WAN interface in the pfSense webConfigurator.

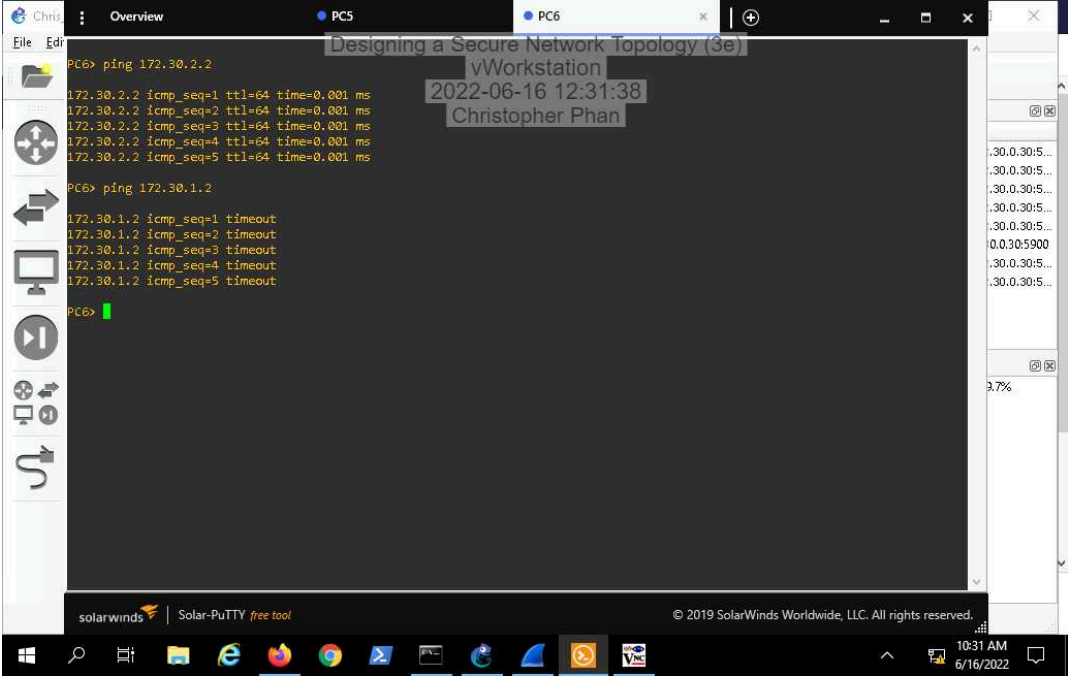


Part 2: Validate DMZ Connectivity

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Make a screen capture showing the results of both pings.



The screenshot shows a SolarWinds vWorkstation terminal window titled "Overview" with tabs for "PC5" and "PC6". The terminal displays the results of two ping commands executed from PC6. The first command is "ping 172.30.2.2", which successfully reaches the destination with five sequential responses, each showing a TTL of 64 and a time of 0.001 ms. The second command is "ping 172.30.1.2", which results in five sequential timeouts. The terminal window includes a left-hand toolbar with various icons for file operations and a right-hand pane showing network statistics. The bottom status bar indicates the software is SolarWinds vWorkstation, version 2022-06-16 12:31:38, and is copyrighted by SolarWinds Worldwide, LLC.

```
PC6> ping 172.30.2.2
172.30.2.2 icmp_seq=1 ttl=64 time=0.001 ms
172.30.2.2 icmp_seq=2 ttl=64 time=0.001 ms
172.30.2.2 icmp_seq=3 ttl=64 time=0.001 ms
172.30.2.2 icmp_seq=4 ttl=64 time=0.001 ms
172.30.2.2 icmp_seq=5 ttl=64 time=0.001 ms

PC6> ping 172.30.1.2
172.30.1.2 icmp_seq=1 timeout
172.30.1.2 icmp_seq=2 timeout
172.30.1.2 icmp_seq=3 timeout
172.30.1.2 icmp_seq=4 timeout
172.30.1.2 icmp_seq=5 timeout

PC6>
```